0 Start here

FULL

00b. Before you start: what you need, what to sign up for, and how long it takes

No surprises: everything you will need is on this page, sorted into **necessary** and **optional**, with how long each part usually takes. Nothing has to be done in one go. Every step can be stopped and carried on later, and the menu remembers where you were.

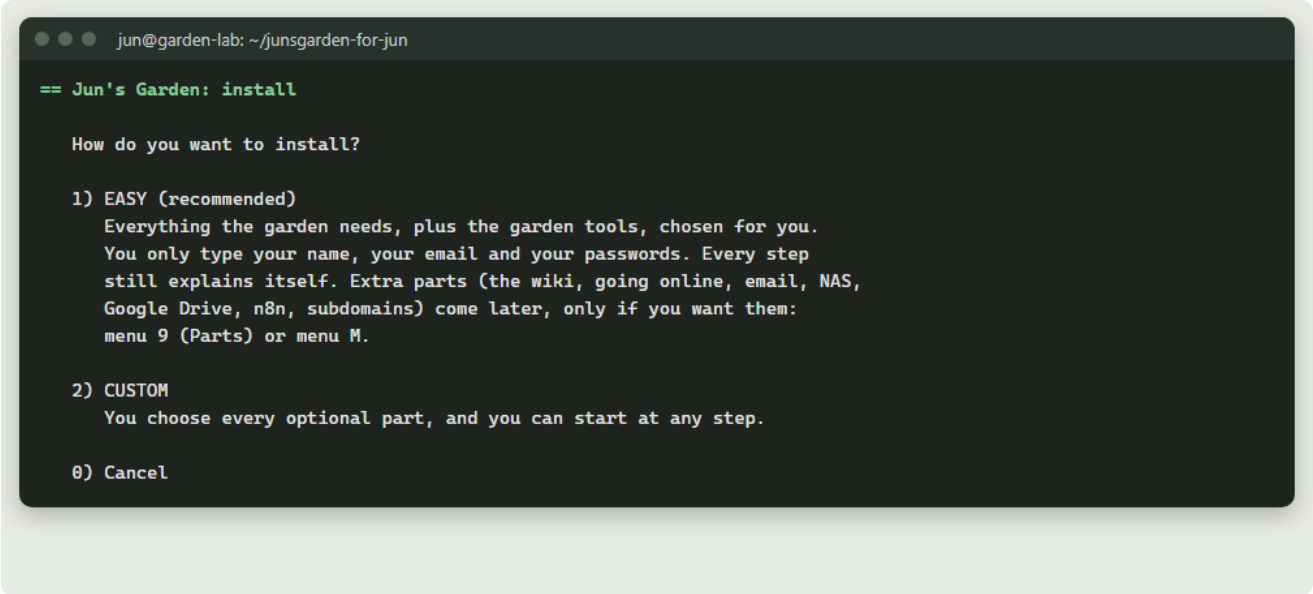
Why it is built this way (a note from Jason)

I chose the automatic route so you can have your own Linux web server without the steep learning curve. The setup does the work and explains every step as it goes, so it stays easy, and Custom lets you choose every part yourself. It is designed to be done in parts: stop, come back, carry on. Later you can move the main folders (menu 8) and choose your own web name (the Website name swap).

It is built to protect your home network and your computer. Only the front door (Caddy) is reachable from the internet, and only on the two web ports. Everything else listens on the garden computer alone. The firewall is on, nobody has a built-in account, and every password and key is made new on your computer. I followed every best practice I could within a home budget, to keep what is exposed to the internet as small as possible.

This is now your project.

Easy or Custom?



The install asks this once. Both use the same 14 numbered steps.

	1) EASY (recommended)	**2) CUSTOM**
Questions	Your name, email, database password (Enter = made for you), keep the web name or choose a new one. At the end: your password and your AI keys	The same, plus each optional part, one by one, and which step to start at
Installs	Everything necessary, plus the garden tools	What you choose: garden tools (default Yes), wiki (default No), n8n (default No)
Photos	Loaded by itself, when the photos download is there	You choose: now, or later (menu 6, then 11)
Wiki	Off. Switch it on any time (menu 9)	Asked (default No)

Either way you can add or remove parts later (guide 19 lists every option).

What you need: necessary

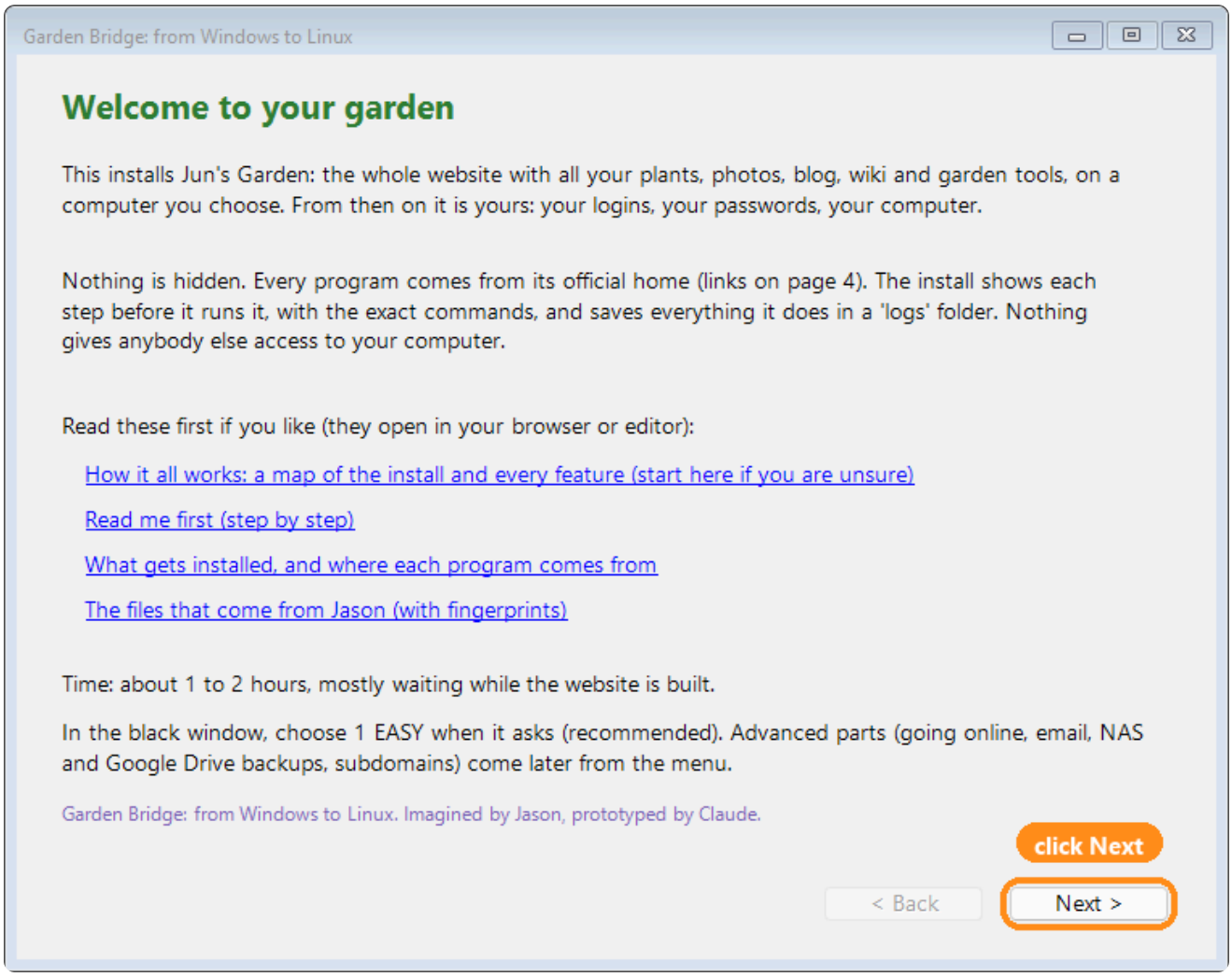
What	Why	Time
A garden computer that can stay on: an Ubuntu laptop (tested), this Windows PC with Windows' own Ubuntu (tested; for using the garden on this PC only, see below), or a Raspberry Pi 5 (tested on 64-bit Raspberry Pi OS; it needs good cooling, a fan, because it gets hot while the website is built)	The garden runs on it	Laptop: 2 to 3 hours to put Ubuntu on it the first time (guide 01). A Pi 5 with its starter kit: about 1 hour. Windows: set up by the installer (one restart)
Free disk space: about 60 GB on an Ubuntu laptop; on Windows, about 60 GB free on C:	Docker, the website, the photos, and the 7 nightly backups (about 6 GB together)	Nothing to do; just check before you start
A USB stick, 16 GB or larger (only to put Ubuntu on a laptop)	To install Ubuntu (it is erased)	Included in the 2 to 3 hours
The three downloads from the Google Drive folder	The garden (about 70 MB), the photos (about 800 MB), the wiki pages (under 1 MB; only if you want the wiki)	Garden: 1 to 2 minutes. Photos: about 2 minutes on fast internet, 15 to 20 minutes on slow internet
The install (menu 1)	Puts everything together	30 to 90 minutes, mostly waiting. Each step shows its own time. The longest step, building the website, takes 15 to 45 minutes (up to an hour on a very old laptop or a Pi)
Your password for the garden	Your login, and your sign-in page	1 minute

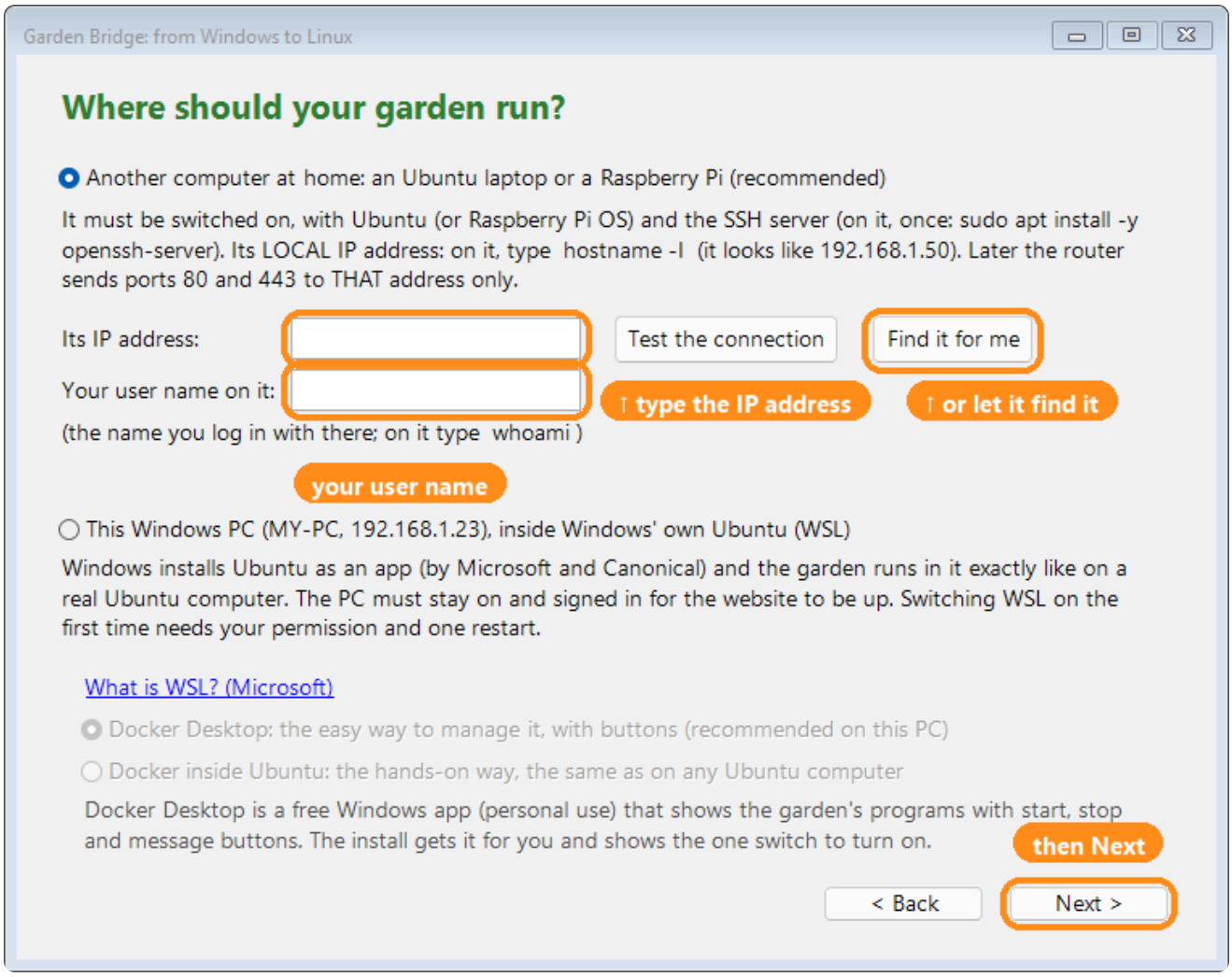
No sign-ups are needed to install. Your garden works at home without any account anywhere.

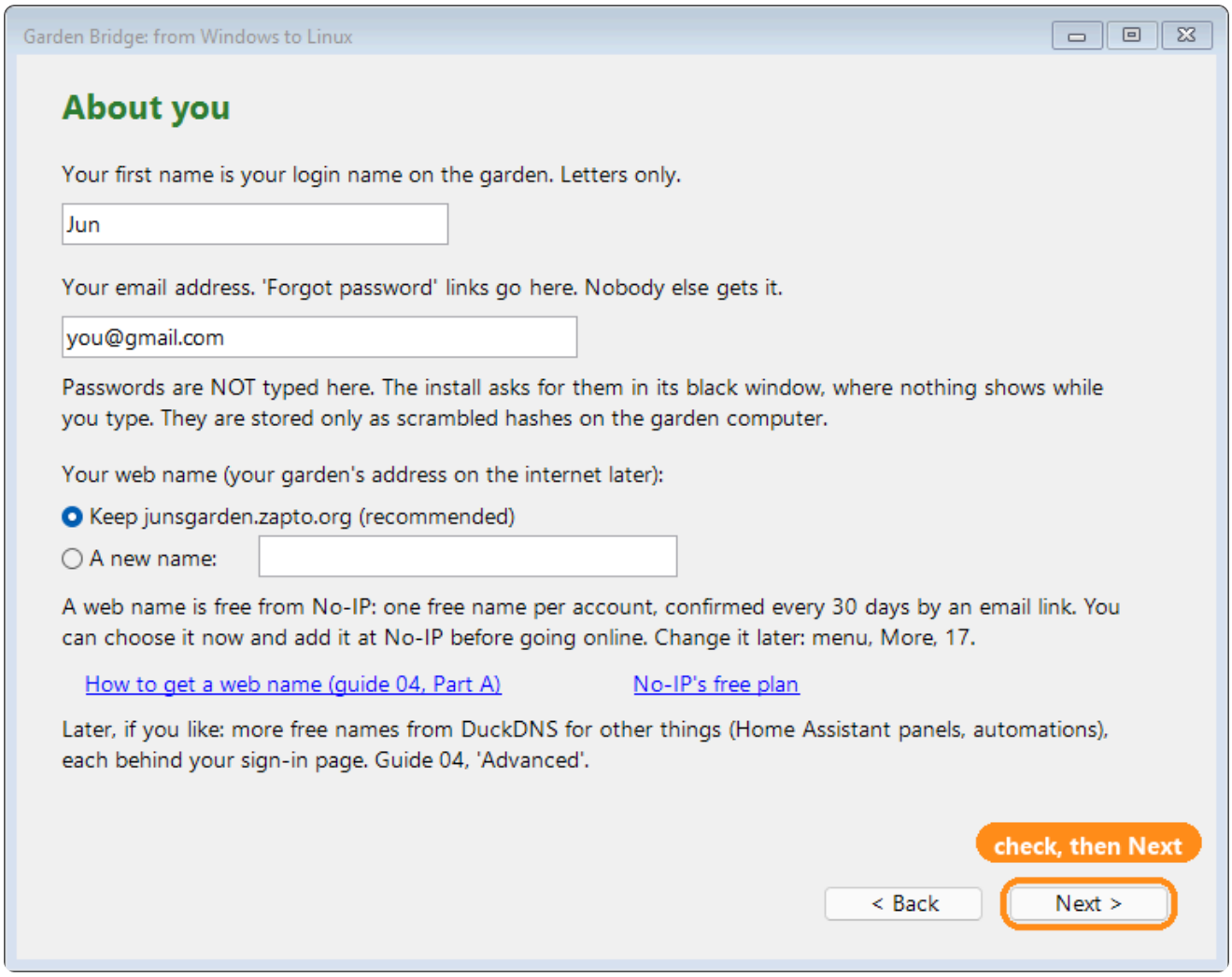
Good to know before the install starts

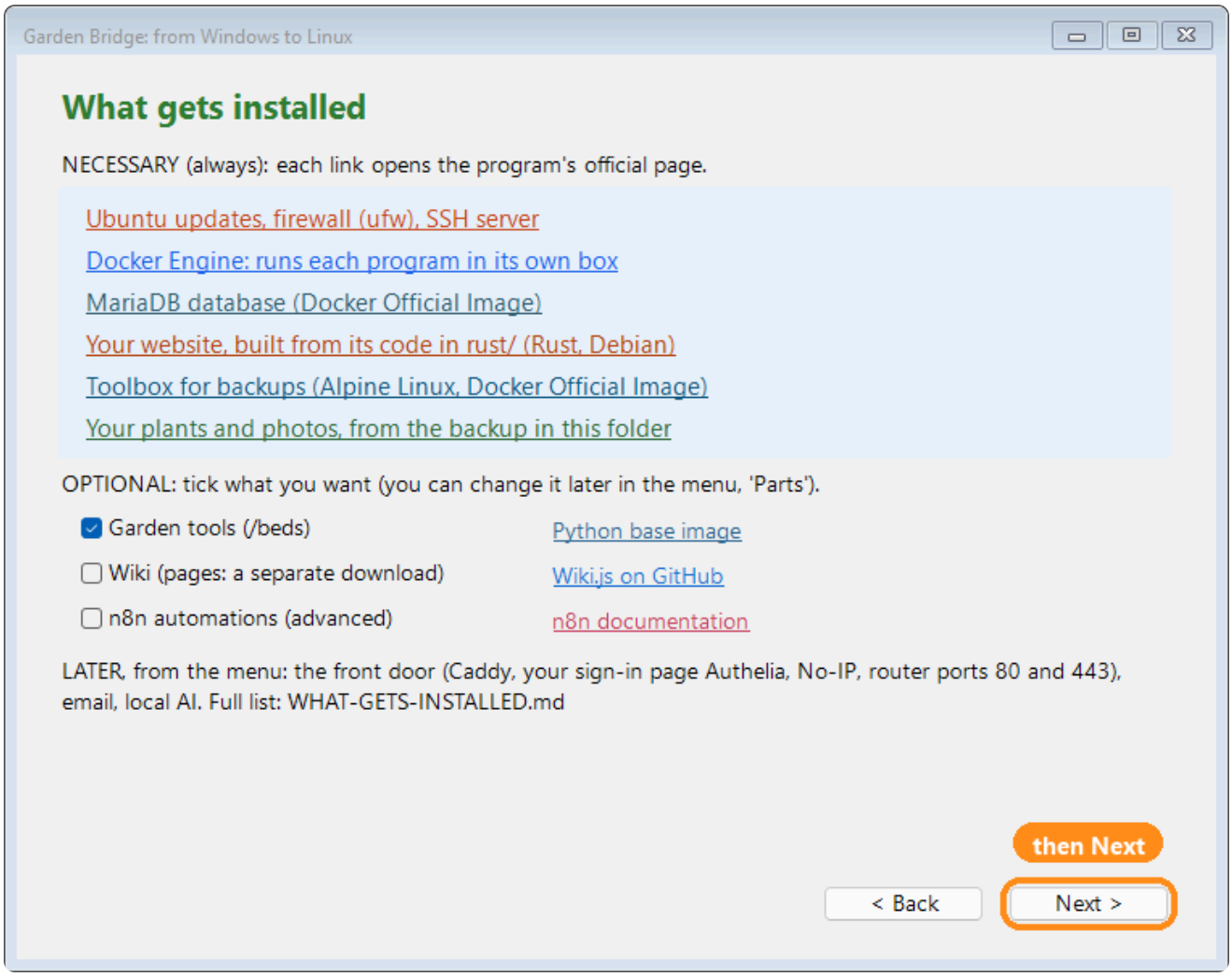
- **Right after a fresh Ubuntu install**, Ubuntu updates itself in the background for 10 to 30 minutes. If step 3 of the install (Ubuntu updates and tools) says it cannot install, wait 15 minutes and choose `1` again: it carries on.
- **After the long build** (building the website) it may ask for your Ubuntu password again. Nothing happens until you type it, so check the window now and then.

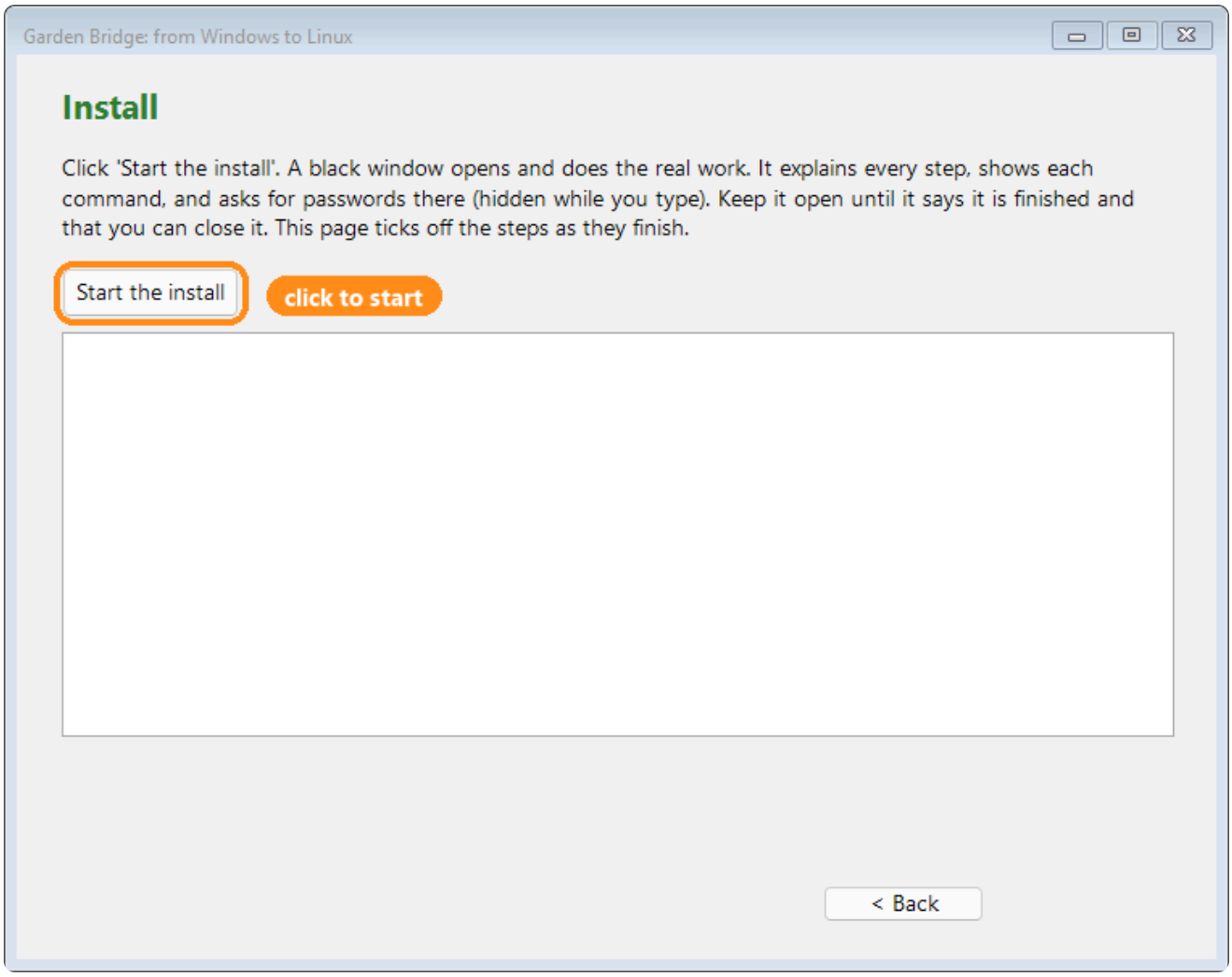
On a Windows PC (the garden inside Windows' own Ubuntu, WSL)




Garden Bridge: from Windows to Linux
Where should your garden run?
Another computer at home: an Ubuntu laptop or a Raspberry Pi (recommended)
It must be switched on, with Ubuntu (or Raspberry Pi OS) and the SSH server (on it, once: sudo apt install -y openssh-server). Its LOCAL IP address: on it, type hostname -I (it looks like 192.168.1.50). Later the router sends ports 80 and 443 to THAT address only.
Its IP address:
Test the connection
Find it for me
Your user name on it:
↑ type the IP address
↑ or let it find it
(the name you log in with there; on it type whoami)
your user name
This Windows PC (MY-PC, 192.168.1.23), inside Windows' own Ubuntu (WSL)
Windows installs Ubuntu as an app (by Microsoft and Canonical) and the garden runs in it exactly like on a real Ubuntu computer. The PC must stay on and signed in for the website to be up. Switching WSL on the first time needs your permission and one restart.
What is WSL? (Microsoft)
Docker Desktop: the easy way to manage it, with buttons (recommended on this PC)
Docker inside Ubuntu: the hands-on way, the same as on any Ubuntu computer
Docker Desktop is a free Windows app (personal use) that shows the garden's programs with start, stop and message buttons. The install gets it for you and shows the one switch to turn on.
then Next
< Back
Next >


Garden Bridge: from Windows to Linux
About you
Your first name is your login name on the garden. Letters only.
Jun
Your email address. 'Forgot password' links go here. Nobody else gets it.
you@gmail.com
Passwords are NOT typed here. The install asks for them in its black window, where nothing shows while you type. They are stored only as scrambled hashes on the garden computer.
Your web name (your garden's address on the internet later):
Keep junsgarden.zapto.org (recommended)
A new name:
A web name is free from No-IP: one free name per account, confirmed every 30 days by an email link. You can choose it now and add it at No-IP before going online. Change it later: menu, More, 17.
How to get a web name (guide 04, Part A)
No-IP's free plan
Later, if you like: more free names from DuckDNS for other things (Home Assistant panels, automations), each behind your sign-in page. Guide 04, 'Advanced'.
check, then Next
< Back
Next >


Garden Bridge: from Windows to Linux
What gets installed
NECESSARY (always): each link opens the program's official page.
Ubuntu updates, firewall (ufw), SSH server
Docker Engine: runs each program in its own box
MariaDB database (Docker Official Image)
Your website, built from its code in rust/ (Rust, Debian)
Toolbox for backups (Alpine Linux, Docker Official Image)
Your plants and photos, from the backup in this folder
OPTIONAL: tick what you want (you can change it later in the menu, 'Parts').
Garden tools (/beds)
Python base image
Wiki (pages: a separate download)
Wiki.js on GitHub
n8n automations (advanced)
n8n documentation
LATER, from the menu: the front door (Caddy, your sign-in page Authelia, No-IP, router ports 80 and 443), email, local AI. Full list: WHAT-GETS-INSTALLED.md
then Next
< Back
Next >



- **What it is for:** using the garden on this PC itself. Putting the garden on the internet, and showing it to phones and other computers at home, does not work from Windows' own Ubuntu as the guides describe: it has its own internal address inside the PC that the router and your phones cannot reach. For those, use a separate garden computer (an Ubuntu laptop or a Raspberry Pi). To learn more, Microsoft explains WSL's networking here (reading only; the garden does not set it up): https://learn.microsoft.com/windows/wsl/networking
- **Which Docker (page 2): Docker Desktop**, the easy way: a Windows app with buttons that shows the garden's programs, starts with Windows, and is free for personal use. The installer gets it for you and waits for one switch (Settings > Resources > WSL

integration: `JunsGarden` on). Or **Docker inside Ubuntu**, the hands-on way: the same as on any Ubuntu computer. Guide 26 compares them.

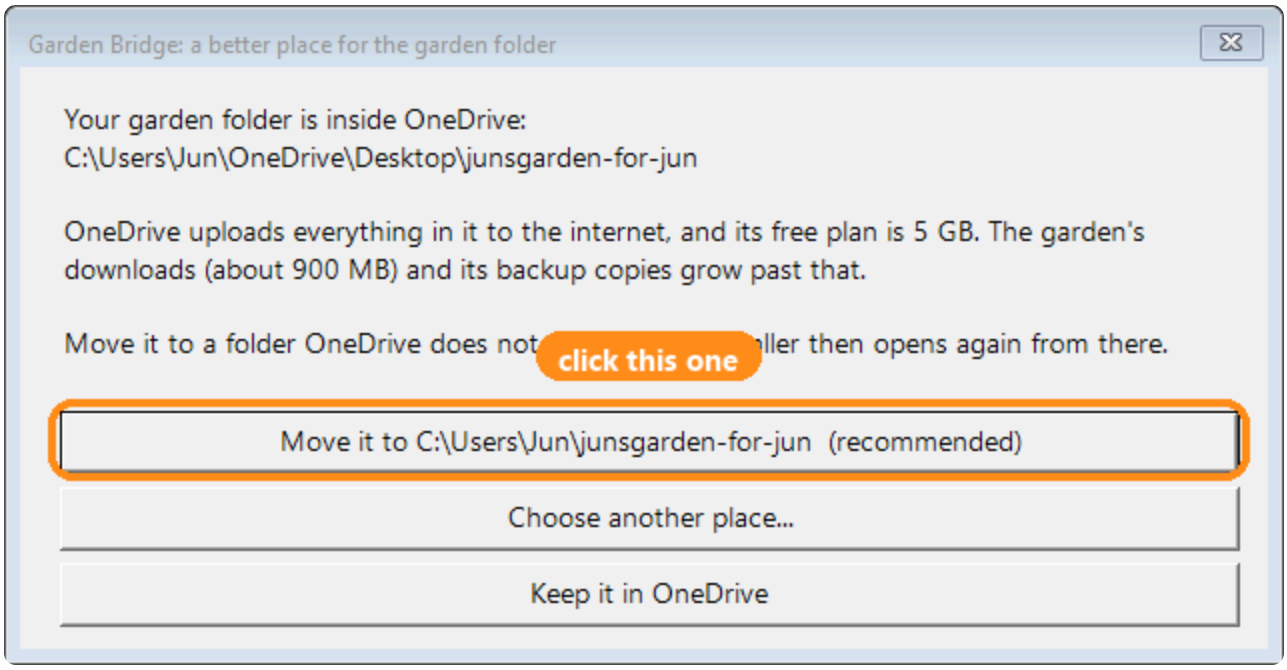
- **One restart:** the first time, Windows asks permission to switch WSL on and needs one restart. After the restart, double-click `Start-Here-Windows.cmd` again and click through the same pages: it carries on where it stopped.
- **Your Ubuntu user name and password:** it asks you to choose an Ubuntu user name (for example `jun`) and a password, typed twice (nothing shows while you type). Write them down. It asks for that password again right after.
- **Download:** Ubuntu for WSL downloads several hundred MB.
- **If WSL does not start after the restart:** virtualization may need to be switched on in the PC's BIOS. Microsoft's help: https://learn.microsoft.com/windows/wsl/troubleshooting
- **The PC must not sleep**, or the garden stops: Settings, System, Power, Sleep = **Never**.
- **About 60 GB free on C:** (see the table above).

If your Desktop is in OneDrive

On many Windows 11 PCs the Desktop, Documents and Pictures folders live inside **OneDrive**. OneDrive is Microsoft's online storage: it uploads (copies to the internet) everything in its folders. You can tell by the folder's path in File Explorer: it has `OneDrive` in it, for example `C:\Users\Jun\OneDrive\Desktop\junsgarden-for-jun`.

That is a problem for the garden. OneDrive's free plan holds **5 GB**. The garden's downloads are about 900 MB, and its backup copies grow past 5 GB, so OneDrive soon fills up and keeps uploading large files for nothing.

So when you double-click `Start-Here-Windows.cmd` and the garden folder is inside OneDrive, a small window opens first, before the installer's pages:



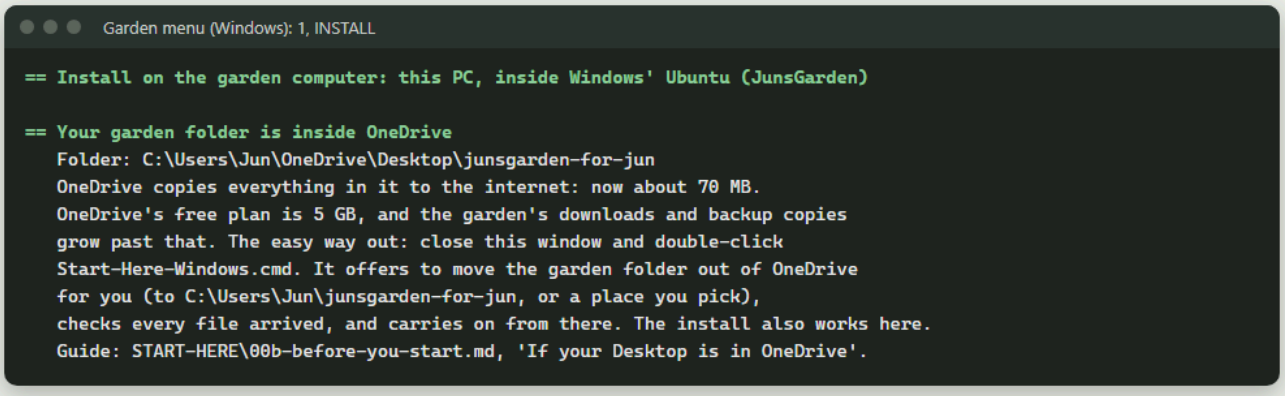
- **Move it to `C:\Users\<you>\junsgarden-for-jun` (recommended):** the garden folder goes into your own user folder (`<you>` is your Windows user name), which OneDrive does not upload. Choose this one.
- **Choose another place...:** a folder window opens. A `junsgarden-for-jun` folder is made inside the place you pick. A place inside OneDrive is refused.
- **Keep it in OneDrive:** nothing moves, and the installer opens as usual. Closing the small window does the same.

What **Move** does, step by step: 1. It copies every file into the new folder. If that folder already has something in it, it stops and moves nothing. 2. It counts the files in both places to check every one arrived. If any are missing, it says so and the old folder stays as it was. 3. If the photos and wiki zips (`junsgarden-photos-addon.zip`, `junsgarden-wiki-addon.zip`) are on a OneDrive Desktop, it moves them into the new folder too. 4. If you had already set up the garden's automatic jobs on this PC (the weekly backup copy, "keep running"), it points them at the new folder, so they keep working after you delete the old one. 5. It puts a **Garden folder** shortcut on your Desktop, so the new folder is easy to find. 6. It says "Done" and opens the installer again from the new place. Carry on there.

The old folder is never deleted for you. It stays in OneDrive until you delete it yourself. After the install has finished, right-click the old `junsgarden-for-jun` folder in OneDrive and choose **Delete** to free the space. (The **Garden folder** shortcut leads to the new one, which you keep.)

> **Why let the installer move it, not File Explorer?** Dragging a folder by hand moves the files one by one, with no check at the end. If it stops halfway, some files are in each place and nothing tells you. The installer copies instead: it checks that every file arrived before it opens from the new place, and it leaves the old folder untouched. If anything goes wrong, the old folder is still complete, so you can simply carry on with it.

The Windows menu (`Garden-Menu-Windows.cmd`) reminds you too. If the garden folder is inside OneDrive, **INSTALL** (menu `1`) shows how much is there and points back to this section. The install also works from OneDrive; the note is only a warning:



The photos and wiki zips are found in the garden folder, on the Desktop (also a OneDrive Desktop) or in Downloads. The install sends them straight to the garden computer from there; no extra copy is made on the PC.

To put the garden on the internet: necessary for that step only

This needs a separate garden computer (an Ubuntu laptop or a Raspberry Pi), not Windows' own Ubuntu on a Windows PC (see "On a Windows PC" above).

What	Why	Time
A free No-IP account with your web name, and its **DDNS key** (guide 04, Part A)	Keeps your web name pointing at your home	About 10 minutes, plus confirming the email No-IP sends. A free name is confirmed every 30 days (No-IP emails a link)
Your router's admin page and its password (on a sticker on the router, or from your internet company)	To reserve the garden computer's address and forward ports 80 and 443 to it (guide 14)	15 to 30 minutes
The front door step (menu M, then 1)	Caddy, your sign-in page, the No-IP updater	15 to 30 minutes

Strongly recommended

What	Why	Time
A Gmail just for your website (recommended, free): https://accounts.google.com/signup	One account for everything the website needs: it sends the password emails, keeps the Google Drive backups and the AI key, and signs in to Tailscale, DuckDNS and No-IP (they offer "Sign in with Google"). Your personal inbox stays apart, and handing the website on later means handing on one account	About 10 minutes

What	Why	Time
A Gmail app password (needs 2-Step Verification on your Google account)	So "forgot password" emails reach you. You only need to remember one password; if you forget it, a reset email comes straight to you (the website and the sign-in page each send their own; menu U then 5 sets both to the same new one again)	5 minutes (10 if 2-Step Verification is not on yet)
A password manager, or a notebook kept somewhere safe	For the keys and passwords below	5 minutes

Plan your website's name and email (10 minutes, on paper)

1. A name. Short, easy to say on the phone, lowercase letters, numbers and `-` only, for example `momsgarden`. It becomes your web name (`momsgarden.zapto.org`, guide 04) and, if it is free, your website's Gmail (`momsgarden.site@gmail.com`). Check both are free before you decide.

2. An email. Three choices; each works:

Choice	Good	Less good
Your own email	Nothing new to make	The website's mail mixes with yours; handing the website on later is harder
One you already use for sign-ups (or spam)	Keeps your main inbox quiet	Easy to miss the important mails (password links, No-IP's 30-day email)
A new Gmail just for the website (recommended): https://accounts.google.com/signup	Everything in one place, apart from your own mail; easy to hand on	One more password (keep it in your password manager)

3. One Gmail, many addresses ("plus addresses"). Gmail delivers `momsgarden.site+noip@gmail.com` to `momsgarden.site@gmail.com`: add `+` and any word before the `@`. Give each service its own word when you sign up:

Service	Sign up with
No-IP (web name)	`momsgarden.site+noip@gmail.com`
DuckDNS, Tailscale	`momsgarden.site+duckdns@gmail.com`, `momsgarden.site+tailscale@gmail.com`
Google AI Studio, OpenRouter	`momsgarden.site+ai@gmail.com`

The "To" line of each email then shows which service it came through, and spam to `+noip` tells you who passed your address on. To sort them for you: in Gmail, the search box, **Show search options**, type the address under **To**, **Create filter**, then choose a label (for example "No-IP"). Google's steps: https://support.google.com/a/users/answer/9308648 and https://support.google.com/mail/answer/6579.

Good to know: - A few websites refuse a `+` in an address; then use the plain one. - Someone can remove the `+word`, so it tells you where an address leaked, but does not hide it. - Dots do not matter in Gmail (`momsgarden.site` = `momsgardensite`): https://support.google.com/mail/answer/7436150. - The Gmail app password (menu `M`, `3`) belongs to the plain address: type that one there. - The garden's own sign-in is always your garden password, never this email account.

Optional (only if you want them)

What	What it gives you	Time
A **Gemini key** (Google AI Studio, free)	The AI boxes: reading plant tags, plant facts (guide 16)	2 minutes
An **OpenRouter key** (free)	A backup for when Gemini is busy	5 minutes

What	What it gives you	Time
A **Perenual key** (free)	Plant facts in the plant admin	5 minutes
The **wiki** (menu 9)	Your garden encyclopedia	5 minutes, plus about 30 minutes while the pages go in
DuckDNS (free)	More addresses for other things, such as Home Assistant (guide 04, Advanced)	5 minutes
A **NAS** (network drive)	A second copy of every backup (guide 13)	10 to 20 minutes
Google Drive backups with your own Google client ID	A copy away from home (guide 13)	15 minutes
SSH keys and Remote Desktop	Work on the garden computer from another room (guide 20)	10 to 20 minutes
Local AI on a desktop with a graphics card	AI without the internet (guide 15)	30 to 60 minutes

Write these down (keep them in a password manager)

What	Where you find it
The garden computer's **user name** and its **password**	You choose them when installing Ubuntu (or, on Windows, when the installer sets up Windows' own Ubuntu). Explained in START-HERE/00-READ-ME-FIRST.md, section "Three names for one computer"
Its **computer name** (host name)	Ubuntu: Settings, System, About ("Device Name"). Also shown by menu M, then 2 (Network)
Its **local IP address** (looks like 192.168.1.50)	Ubuntu: Settings, Wi-Fi (or Network), the gear next to your connection, "IPv4 Address". Or menu M, then 2. Or on a Windows PC: the installer's **Find it for me** button

What	Where you find it
Your **router's address and password**	The sticker on the router
Your **garden password**	You choose it at the end of the install
The **wiki admin password** (only with the wiki)	You choose it when the wiki is set up
Your **No-IP DDNS key** (username and password)	No-IP, Dynamic DNS, DDNS Keys
Your **Gmail app password**	myaccount.google.com, Security, App passwords
Your **AI keys**	Their own pages (guide 16)

Ubuntu has no separate "root password": when the computer asks for a password to make a change (`sudo`), it means **your own** Ubuntu password.

Ports: what is opened, and where

Port	What	Open to
80 and 443	The front door (web)	The internet, only after the front door step, and only these two in the router
22	SSH (working from another computer)	Your home network only
3389	Remote Desktop, if you switch it on (guide 20)	Your home network only; never forwarded in the router
Everything else (8030 website, 3000 wiki, 9091 sign-in page, the database, ...)	The garden's programs	This computer only

A realistic plan

Sitting	What	Time
1	Put Ubuntu on the laptop (guide 01), or set up the Pi	1 to 3 hours
2	Download, then install (Easy)	1 to 2 hours, mostly waiting (the website build alone: 15 to 45 minutes, up to an hour on a very old laptop or a Pi)
3	Gmail app password, AI keys (menu K)	15 minutes
4	No-IP, router, front door: the garden goes online	About 1 hour
Later	Anything optional, one at a time	See the table above

If you stop in the middle of a sitting, start the menu again and choose `1`: it shows which steps are done and carries on.